PO12Q - Introduction to Quantitative Political Analysis II

Exam 2025/26

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Paper Code: PO12Q –
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Exam Rubric

Time Allowed: 2 Hours
Exam Type: Written - Standard
Approved Calculators: Permitted

UNIVERSITY OF WARWICK

Table of Contents

Instructions

- Part I contains the questions for the exam
  - These are divided into four blocks: A, B, C, and D.
  - Block A is compulsory and must be answered by all candidates.
  - You must then select ONE block out of B, C, and D.
  - Should you change your mind during the exam, indicate clearly which block you wish to have marked.
  - You can achieve a maximum of 120 points.
  - Information on the maximum number of points obtainable is outlined at the beginning of each block
- Part II contains the case study to which all questions in Block A of Part I refer
- Part III contains the relevant statistical tables
  - Normal curve tail probabilities are provided in Table 5, page 14.
  - t-distribution critical values are provided in Table 6, page 15.
  - χ^2-distribution critical values are provided in Table 7, page 16.
- Part IV contains the formula collection and notation of the module
  - If you use notation other than in the formula collection (page 17) or notation table (page 19), give an explanation.
- Allowed time: 2 hours.
- A university-approved calculator is allowed.

Part I: Exam Questions

1 | Block A – Compulsory

80 points, points for each question in parentheses

All questions in this section relate to the case study in Part II.

Level 1 (each question 4 points)

1. Interpret the output in Subsection 12.1.
2. Table 4: State the null- and alternative hypotheses underpinning Model 1.

Level 2 (each question 6 points)

3. Subsection 11.1: Explain (show the calculations) how the χ^2 value is obtained.
4. Table 4: Interpret the intercept in Model 1.
5. Table 4: Interpret the coefficient of "Median household income (£)" in Model 4.
6. Table 4: Interpret the model fit for Model 5.
7. Which of the regression models shown in Table 4 is the best model overall? Explain your reasoning.
8. Why is the relationship between household income and house price statistically significant in Model 1 (see Table 4), but insignificant in the crosstabulation (Subsection 11.1)?
9. Does the output in Subsection 14.1 allow you to make any statements about $\text{cov}(\boldsymbol{X}, \boldsymbol{\epsilon})$? Why / why not?
10. Subsection 14.3: Interpret the test.

Block A is continued on page 5.

Level 3 (each question 8 points)

11. Compare the output of Subsection 14.2 with the regression results in Table 4. What do you notice, and what might explain the difference?

12. Consider Figure 1. Assess which of the Classical Linear Assumptions are satisfied, which ones are violated, and which ones can we not make a statement about on the basis of the plot. Explain your judgement.

13. Consider the `modelsummary` code below. In conjunction with the previous code chunks in this case study, would this code produce Table 4? Explain your answer.

```
library(modelsummary)

models <- list(
  "(1)" = model1,
  "(2)" = model2,
  "(3)" = model3,
  "(4)" = model4,
  "(5)" = model5)

cm <- c('income' = "Median household income (£)",
        'crime_cathigh' = "Crime Category (High)",
        '(Intercept)' = '(Intercept)')

modelsummary(models,
             title = 'Regression Models',
             gof_omit = 'DF|Deviance|Log.Lik|F|AIC|BIC|RMSE|R2',
             stars=TRUE),
  group_tt(j = list("\\makecell{Dependent Variable:\\\\Median house price (£)}" = 2:6))
```

Bonus (4 points, you can achieve a maximum of 120 points)

- Which is Flo's favourite Stephen King novel?

2 | Block B – True / False Statements

40 points, 5 points for each statement

State whether each of the following statements is true or false. Explain why the statement is true or false.

1. The chi-square statistic increases as the sample size grows, even if the degree of statistical dependence remains constant.
2. Adding a variable to an Ordinary Least Squares (OLS) model can reduce the value of R^2.
3. In the case of perfect (multi-)collinearity, $\boldsymbol{X'X}$ is singular.
4. In a two-sample test for means, the standard error of the difference in means decreases if both sample means are closer together.
5. In regression analysis, the principle of ceteris paribus only applies to those variables included in the model.
6. The intercept in a regression model always has a meaningful interpretation.
7. In OLS, the regression line passes through the point $(\bar{x}, \bar{y})$.
8. In two-sample tests, the pooled standard error is used when the variances of the two groups are assumed to be equal.

3 | Block C – Calculations

40 points, points for each question in parentheses

Table 1 shows data for a sample of five fictitious countries. It contains information about the military spending and diplomatic influence score on the international stage (measured on a 0–100 scale from expert surveys).

1. Specify the null and alternative hypothesis for the relationship between military spending and the diplomatic influence score of a country. **(4)**
2. Fit a linear regression model of the type $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ to the data in Table 1. Specify the full sample regression function. **(18)**
3. Determine whether we can reject the null hypothesis at a 95% confidence level. **(18)**

Country	Military Spending (% GDP)	Diplomatic Influence Score
A	2.5	68.2
B	3.8	74.0
C	1.9	63.5
D	4.2	79.1
E	2.0	66.3

Table 1: Military Spending and Diplomatic Influence by Country

4 | Block D – Explanations

40 points, 8 points for each method / statement

Explain each of the following methods or concepts in your own words, and illustrate your explanation with a worked example of your choice.

1. Conditional Expectation Function
2. Determinant
3. (Multi-)Collinearity
4. Omitted Variable Bias
5. Variance-Covariance Matrix

Part II: Case Study

5 | Preface

The data set for this case study contains several variables for each of the London wards, describing, amongst other things, demographics of their population and the number of crimes committed in each of them in 2015. A full codebook is provided in Table 2. The data are taken from London Data Store, 2013.

6 | Code Book

Variable	Label	Year
ward	Ward name	n/a
inner	Binary classification of whether the ward is inner or outer London (based on ONS classification)	n/a
area	Land area (KM^2)	n/a
population	Population	2015
children	Number of people aged 0-15	2015
adults	Number of people aged 16-64	2015
elderly	Number of people aged 65+	2015
age	Mean age of population	2013
education	Percentage of population with level 4 qualifications and above	2011
crime	Crimes committed per 1,000 residents	2015
employed	Number of people aged 16-64 in employment	2015
benefits	Percentage of population claiming work-related benefits	2011
migration	Net rate of worker-aged migration	2012
income	Median household income (£)	2013
houseprice	Median house price (£)	2014
cars	Average number of cars per household	2011
turnout	Turnout at the 2012 mayoral election (%)	2012

Table 2: Codebook for P012Q_exam Data Set

7 | Preliminaries

```
london <- read.csv("P012Q_exam.csv")

library(tidyverse)
```

8 | Descriptives for `london` Data Frame

	N	Mean	SD	Min	P25	Median	P75	Max
area	625	2.55	2.58	0.40	1.20	1.90	2.90	29.00
children	625	2763.92	877.32	650.00	2150.00	2650.00	3300.00	6450.00
adults	625	9478.56	2261.59	3050.00	7750.00	9400.00	10 950.00	21 750.00
elderly	625	1571.52	523.55	600.00	1200.00	1450.00	1800.00	3600.00
age	625	35.96	3.07	28.70	33.70	35.50	38.00	44.30
education	625	37.66	12.86	12.50	27.30	35.50	47.00	68.70
crime	625	83.07	74.41	24.50	54.50	70.40	90.80	1212.10
employed	625	6248.33	1330.35	2293.00	5319.00	6103.00	7038.00	13 759.00
benefits	625	12.32	5.10	0.50	8.10	12.20	16.10	29.50
migration	625	44.04	30.52	0.70	20.00	38.90	60.50	149.20
income	625	39 263.70	7454.24	25 090.00	33 600.00	38 200.00	43 470.00	88 330.00
houseprice	625	434 450.74	268 205.63	173 000.00	280 000.00	363 500.00	485 000.00	3 500 000.00
cars	625	0.84	0.33	0.20	0.60	0.80	1.10	1.70
turnout	625	34.16	5.36	19.30	30.90	34.20	37.60	51.70

Table 3: Descriptive Statistics for 'london' Data Frame

9 | Proportion of missing values per variable

```
##       ward    borough      inner       area   children     adults    elderly
##          0          0          0          0          0          0          0
##        age  education      crime   employed   benefits  migration     income
##          0          0          0          0          0          0          0
## houseprice       cars    turnout
##          0          0          0
```

10 | Recoding

10.1 houseprice

```
london <- london %>%
  mutate(houseprice_cat=
             cut(houseprice, breaks=c(0,363500,3500001),
                 labels=c("low","high")))
```

10.2 crime

```
london <- london %>%
  mutate(crime_cat=
             cut(crime, breaks=c(0,quantile(london$crime, probs = 0.4),1213),
                 labels=c("low","high")))
```

11 | Crosstabulations

11.1 Houseprice and Crime

```
table1 <- with(london, table(crime_cat, houseprice_cat))
table1
##           houseprice_cat
## crime_cat low high
##      low  127  123
##      high 186  189

Xsq1 <- chisq.test(table1, correct=FALSE)
Xsq1
##
##  Pearson's Chi-squared test
##
## data:  table1
## X-squared = 0.0864, df = 1, p-value = 0.7688
```

12 | Two-Sample Tests

12.1 Levene Test

```
leveneTest(london$houseprice ~ london$crime_cat)
## Levene's Test for Homogeneity of Variance (center = median)
##         Df F value    Pr(>F)
## group    1  18.543 1.929e-05 ***
##        623
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

13 | Regression

13.1 Regression Models

```
model1 <- lm(houseprice ~ income, data=london)

model2 <- lm(houseprice ~ crime, data=london)

model3 <- lm(houseprice ~ crime_cat, data=london)

model4 <- lm(houseprice ~ income + crime, data=london)

model5 <- lm(houseprice ~ income + crime_cat, data=london)
```

13.2 Results Table

	Dependent Variable: Median house price (£)				
	(1)	(2)	(3)	(4)	(5)
Median household income (£)	27.817***			26.937***	29.523***
	(0.914)			(0.865)	(0.851)
Crimes committed per 1,000 residents		1091.754***		791.005***	
		(137.623)		(86.618)	
Crime Category (High)			60 910.413**		142 588.427***
			(21 780.182)		(12 941.887)
(Intercept)	−657 731.365***	343 756.195***	397 904.494***	−688 887.665***	−810 291.085***
	(36 539.908)	(15 343.010)	(16 870.856)	(34 508.644)	(36 203.311)
Num.Obs.	625	625	625	625	625
R2	0.598	0.092	0.012	0.645	0.663
R2 Adj.	0.597	0.090	0.011	0.644	0.662

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 4: Regression Models

14 | Testing the Classical Linear Assumptions

14.1 bptest()

```
bptest(model1)
##
##  studentized Breusch-Pagan test
##
## data:  model1
## BP = 106.33, df = 1, p-value < 2.2e-16
```

14.2 coeftest()

```
coeftest(model1, vcov = vcovHC(model1, type="HC3"))
##
## t test of coefficients:
##
##                 Estimate  Std. Error t value  Pr(>|t|)
## (Intercept) -657731.365  112116.824 -5.8665  7.23e-09 ***
## income           27.817       2.962  9.3913 < 2.2e-16 ***
## —
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

14.3 vif()

```
vif(model4)
##   income    crime
## 1.012578 1.012578
```

14.4 Residual vs. x Plot

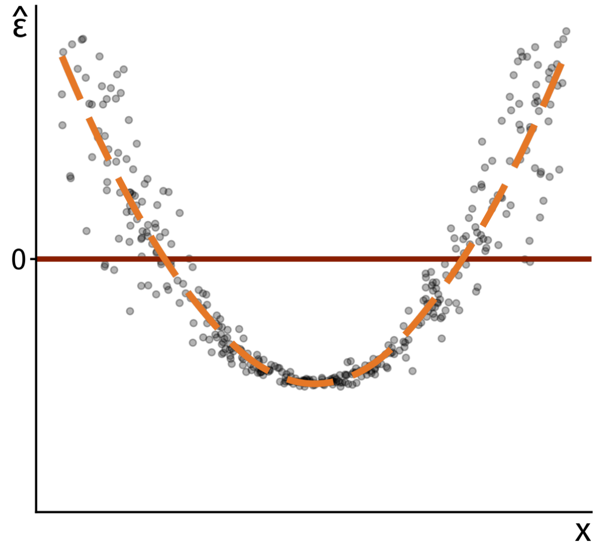
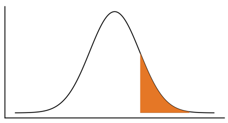


Figure 1: Residual vs. x Plot

Part III: Statistical Tables

15 | Normal-Distribution

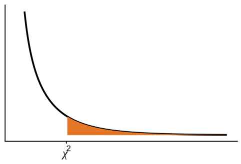


z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002

Table 5: Right-Tail Probabilities under the Normal Distribution

	Confidence Level									
	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	One-Tail Probability									
df	$t_{0.25}$	$t_{0.20}$	$t_{0.15}$	$t_{0.10}$	$t_{0.05}$	$t_{0.025}$	$t_{0.01}$	$t_{0.005}$	$t_{0.001}$	$t_{0.0005}$
1	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.33	31.60
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.22	12.92
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291

Table 6: Probabilities under the t-Distribution



df	0.995	0.99	0.975	0.95	0.9	0.1	0.05	0.025	0.01
	Right-Tail Probability								
1	0.00	0.00	0.00	0.00	0.02	2.71	3.84	5.02	6.63
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14	13.28
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83	15.09
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45	16.81
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.72
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74	27.69
14	4.07	4.66	5.63	6.57	7.79	21.06	23.68	26.12	29.14
15	4.60	5.23	6.26	7.26	8.55	22.31	25.00	27.49	30.58
16	5.14	5.81	6.91	7.96	9.31	23.54	26.30	28.85	32.00
17	5.70	6.41	7.56	8.67	10.09	24.77	27.59	30.19	33.41
18	6.26	7.01	8.23	9.39	10.86	25.99	28.87	31.53	34.81
19	6.84	7.63	8.91	10.12	11.65	27.20	30.14	32.85	36.19
20	7.43	8.26	9.59	10.85	12.44	28.41	31.41	34.17	37.57
22	8.64	9.54	10.98	12.34	14.04	30.81	33.92	36.78	40.29
24	9.89	10.86	12.40	13.85	15.66	33.20	36.42	39.36	42.98
26	11.16	12.20	13.84	15.38	17.29	35.56	38.89	41.92	45.64
28	12.46	13.56	15.31	16.93	18.94	37.92	41.34	44.46	48.28
30	13.79	14.95	16.79	18.49	20.60	40.26	43.77	46.98	50.89
40	20.71	22.16	24.43	26.51	29.05	51.81	55.76	59.34	63.69
50	27.99	29.71	32.36	34.76	37.69	63.17	67.50	71.42	76.15
60	35.53	37.48	40.48	43.19	46.46	74.40	79.08	83.30	88.38
70	43.28	45.44	48.76	51.74	55.33	85.53	90.53	95.02	100.43
80	51.17	53.54	57.15	60.39	64.28	96.58	101.88	106.63	112.33
90	59.20	61.75	65.65	69.13	73.29	107.57	113.15	118.14	124.12
100	67.33	70.06	74.22	77.93	82.36	118.50	124.34	129.56	135.81

Table 7: Probabilities under the χ^2-Distribution

Part IV: Formulae and Notation

18 | Formulae

Statistic	Formula
Crosstabulations	
χ^2-Test Statistic	$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$
Two Sample Tests	
Standard Error of Difference	$se = \sqrt{(se_1)^2 + (se_2)^2}$
Means	
Standard Error of Difference	$se = \sqrt{\frac{se_1^2}{n_1} + \frac{se_2^2}{n_2}}$
Standard Error (pooled)	$se_0 = \sqrt{s_0^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$
Variance (pooled)	$s_0^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$
Proportions	
Standard Error	$se = \sqrt{\frac{\hat{\pi}(1 - \hat{\pi})}{n}}$
Standard Error of Difference	$se = \sqrt{\frac{\hat{\pi}_1(1 - \hat{\pi}_1)}{n_1} + \frac{\hat{\pi}_2(1 - \hat{\pi}_2)}{n_2}}$
Standard Error (pooled)	$se_0 = \sqrt{\bar{\pi}(1 - \bar{\pi})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$
Regression	
Adjusted R^2	$\bar{R}^2 = 1 - \frac{\sum \hat{\epsilon}_i^2 / (n - k - 1)}{\sum (y_i - \bar{y})^2 / (n - 1)}$
Coefficient of Determination	$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum \hat{\epsilon}_i^2}{\sum (y_i - \bar{y})^2}$
Estimated Variance	$\hat{\sigma}^2 = \frac{\sum \hat{\epsilon}_i^2}{n - p}$
Estimator of β_0	$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
Estimator of β_1	$\hat{\beta}_1 = \frac{\sum_{i=1}^{N} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{N} (x_i - \bar{x})^2}$
	$\left.\begin{matrix}\hat{\beta}_0 \\ \hat{\beta}_1\end{matrix}\right\} \hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{y})$
Explained Sum of Squares	$\sum (\hat{y}_i - \bar{y})^2$
Residual Sum of Squares	$\sum \hat{\epsilon}_i^2 = \sum (y_i - \hat{y}_i)^2$
Estimated Standard Error of $\hat{\beta}_0$	$\hat{se}(\hat{\beta}_0) = \sqrt{\frac{\sum x_i^2}{n \sum (x_i - \bar{x})^2}}\ \hat{\sigma}$
Estimated Standard Error of $\hat{\beta}_1$	$\hat{se}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum (x_i - \bar{x})^2}}$
	$\left.\begin{matrix}\hat{se}(\hat{\beta}_0) \\ \hat{se}(\hat{\beta}_1)\end{matrix}\right\} \hat{\mathrm{var}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2 (\boldsymbol{X}'\boldsymbol{X})^{-1}$
Total Sum of Squares	$\sum (y_i - \bar{y})^2$

Table 8: Formulae for PO12Q

18.1 Matrices

$$\boldsymbol{X'X} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}$$

$$(\boldsymbol{X'X})^{-1} = \frac{1}{n \sum x_i^2 - \sum x_i \sum x_i} \begin{bmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{bmatrix}$$

$$\boldsymbol{X'Y} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$$

Symbol	Explanation
χ^2	Chi-Squared for test of independence
π	Population Proportion
$\hat{\pi}$	Sample Proportion
$\bar{\pi}$	Pooled Proportion
se_0	Standard Error under the Null Hypothesis
β	Regression Coefficient
$\hat{\beta}$	Estimated Regression Coefficient
ϵ	Error Term
$\hat{\epsilon}$	Estimated Error Term / Residual
A^{-1}	Inverse of Matrix A
A′	Transpose of Matrix A
I	Identity Matrix
R^2	R-Squared / Model Fit
σ^2	Mean Squared Error
k	Number of Slope Coefficients
$\bar{\mathrm{R}}^2$	Adjusted R-Squared
log(x)	Logarithm of variable x
α_i	Regression coefficients of secondary regression models
p	Total Number of Regression Coefficients, including the Intercept
VcV	Variance-Covariance Matrix
Ω	Is equal to $\sigma^2 I$, where σ^2 represents the mean squared error
t	Time period in time-series data

Table 9: Notation

Part V: References

List of References

London Data Store. (2013). Ward Profiles and Atlas. https://data.london.gov.uk/dataset/ward-profiles-and-atlas